

Input image (camera / upload)



Letterbox $\rightarrow 640 \times 640$



mAP@0.5
0.632

YOLOv8n detection



Crop leaves + $\geq 128^2$ filter



99.5% px
0.915 mIoU

YOLOv8n-seg
(mask · bg $\rightarrow$ black)



Resize $200^2 \rightarrow$ pad 224^2



off by
default

GAN occlusion recovery



89.81%
5-class

MobileNetV3-Small
classifier



Per-class leaf counts
{healthy, blotch, spot, ...}

route by
confidence

conf $\geq \tau_{\text{high}}$ $\rightarrow$
Retrain buffer
(auto-labelled)

conf $< \tau_{\text{low}}$ $\rightarrow$
Relabel queue
(human/LLM review)